



AMAZON EC2 OVERVIEW AND FEATURES

Scalable cloud computing services for flexible infrastructure



AMAZON EC2 FUNDAMENTALS

AMAZON EC2 OVERVIEW



•Scalable Cloud Computing

- Amazon EC2 provides scalable computing power that can be adjusted dynamically based on workload demands.

•On-Demand Virtual Machines

- Users can launch virtual machines called instances on-demand for flexible and rapid deployment.

•Cost Efficiency

- EC2 uses a pay-as-you-go pricing model that helps businesses only pay for the resources they use.

•Integration and Versatility

- EC2 integrates with AWS services and supports diverse use cases like web hosting.

WHAT IS AN EC2 INSTANCE?



Virtual Server Concept

EC2 instances are virtual servers in the cloud offering flexible compute capacity similar to physical servers.

Instance Configuration

Users select instance types defining CPU, memory, storage, and networking; configured via pre-built software templates.

Scalability and Automation

Instances enable quick deployment, scaling, and management using automation tools for efficient cloud computing.

Use Cases

EC2 supports hosting websites, backend apps, data analysis, and enterprise systems with reliable performance.

INSTANCE TYPES



•Instance Type Families

- Instance types are grouped into families like general-purpose, compute-optimized, memory-optimized, and storage-optimized.

•Use Case Specific Design

- Each instance type suits specific workloads, such as CPU-intensive tasks or in-memory analytics for databases.

•Flexibility and Scalability

- Users can select and change instance types to optimize resource use and scale workloads effectively.

•Continuous Optimization

- New instance types with better performance and cost efficiency are regularly introduced by AWS.

KEY FEATURES OF EC2



Virtual Servers and AMIs

EC2 provides virtual servers called instances and preconfigured Amazon Machine Images for simplified deployment.

Instance Types and Storage

Users select instance types for workload needs; storage includes persistent EBS and temporary instance store volumes.

Security and Compliance

Security features include key pairs and security groups; EC2 meets compliance standards like PCI DSS.

Integration and Scalability

EC2 integrates with AWS services to enable monitoring, automation, and scaling of cloud applications.

STORAGE OPTIONS



Persistent Block Storage

EBS provides reliable, persistent storage ideal for databases and operating systems, available even if instances stop.

Temporary Instance Store

Instance store volumes offer fast, temporary storage for caching and intermediate processing but data is lost on stop.

Backup and Recovery Features

EBS supports snapshots enabling data backup and recovery, ensuring data protection in cloud architectures.

Comprehensive Storage Ecosystem

EC2 integrates with other services like object storage, providing flexible and cost-effective storage solutions.

SECURITY COMPONENTS



Secure Access with Key Pairs

Key pairs provide secure login to instances using public and private keys, eliminating passwords and enhancing security.

Security Groups as Firewalls

Security groups act as virtual firewalls controlling inbound and outbound traffic with customizable rules.

Compliance and Identity Management

EC2 supports industry compliance and integrates identity management for strict authentication and authorization.

Monitoring and Auditing Tools

Monitoring and auditing provide visibility into system activities to enhance security and regulatory compliance.

RELATED AWS SERVICES



Auto Scaling for Efficiency

Auto Scaling adjusts EC2 instances automatically based on demand for optimal performance and resource use.

Monitoring with CloudWatch

CloudWatch monitors performance metrics and system health to provide valuable operational insights.

Load Balancing for Availability

Elastic Load Balancing distributes traffic across multiple instances to enhance availability and fault tolerance.

Security and Backup Services

GuardDuty detects threats, while AWS Backup ensures data protection and recovery for EC2 instances.

ADDITIONAL COMPUTE SERVICES



Amazon Lightsail Simplicity

Lightsail offers an easy platform for beginners to build websites and apps with predictable pricing and simple setup.

Elastic Container Service (ECS)

ECS enables deployment and scaling of containerized applications, ideal for microservices architectures.

Elastic Kubernetes Service (EKS)

EKS provides a managed Kubernetes environment for efficient container orchestration on cloud infrastructure.

Tailored Compute Options

Various compute services allow organizations to align infrastructure with expertise and workload needs.

ACCESSING AMAZON EC2

Web-Based Management Console

The AWS Management Console provides a user-friendly web interface for creating and managing EC2 instances visually.

Command Line Interface (CLI)

AWS CLI allows terminal-based control of EC2, ideal for automation and complex scripting tasks.

Infrastructure as Code with CloudFormation

CloudFormation uses JSON or YAML templates to automate provisioning and management of EC2 resources.

Programmatic Access via SDKs and APIs

AWS SDKs and EC2 Query API enable developers to integrate and automate EC2 operations within applications.



EC2 PRICING MODELS



Flexible Pricing Options

EC2 provides varied pricing models to match different usage needs and budgets, ensuring flexibility and cost efficiency.

Commitment-Based Savings

Savings Plans and Reserved Instances offer discounted rates for users committing to consistent long-term usage.

Cost-Effective Spot Instances

Spot Instances allow users to utilize unused cloud capacity at reduced prices with variable availability.

Exclusive Resource Allocation

Dedicated Hosts and Capacity Reservations provide users exclusive physical resources for compliance and performance.

COST OPTIMIZATION AND BILLING



Cost Estimation Tools

AWS Pricing Calculator estimates costs based on expected usage for better budgeting and planning.

Usage Monitoring

Billing dashboard and CloudWatch provide insights into usage patterns and resource utilization.

Cost Reduction Strategies

Rightsizing, Reserved Instances, Savings Plans, Spot Instances, and auto-scaling help reduce costs effectively.

Continuous Optimization

Regular analysis of usage reports and cost trends supports ongoing infrastructure optimization.